

A Multivariate Permutation Test for Correlated Menstrual Cycle

Regularity Counts:

Application to Jewish Halachic Anticipatory Rules

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The dataset analyzed in this study was originally collected by Fehring et al. [6] and is publicly archived at Marquette University [7]. The author declares no conflicts of interest. Analysis code is publicly available at <https://github.com/dvirross/VessetVectorTest>.

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Abstract

Jewish religious law (*Halacha*) prescribes that women anticipate menstruation based on recurring patterns in their cycle history. These anticipatory rules are the *halachic vessel rules* (*vessel*: Hebrew for menstruation). Five testable vessel rule types involve cycle-length data: *Haflaga* (fixed interval), *Dilug* (arithmetic progression), *Week* (weekly anchor), *Week-Dilug* (30-day anchor), and *Dilug-in-Dilug* (second-order progression). These form a polynomial hierarchy with algebraic containment and approximation relationships; pairwise rate-based Spearman correlations confirm inter-pattern association ($\text{Haflaga} \times \text{Dilug-in-Dilug}$: $r_s = 0.27$, Holm-adj $p = .030$). We apply three complementary null models to 1,554 menstrual cycles from 118 women [6]: a global permutation null, a multinomial sampling null, and a within-woman permutation null conditioning on each woman's observed cycle-length multiset. Under the global null, *Haflaga* is strongly overrepresented ($z = 4.51$, $p < .001$); the focused 3-pattern Mahalanobis test detects significant multivariate departure ($D_M = 4.52$, $p < .001$), reflecting concentration of patterns among women with lower cycle-length variability. Under the within-woman null — the scientifically sharper test — no pattern is significant (all Holm-corrected $p \geq .198$; joint $D_M = 2.95$, $p = .121$), and all observed counts fall *below* their within-woman null means; the sample is underpowered for individual within-woman tests. These complementary results suggest that halachic vessel rules function primarily as empirical detectors of individual cycle regularity, with no detectable evidence, under within-woman exchangeability, that temporal ordering adds structure beyond each woman's own cycle-length distribution.

Keywords: menstrual cycle, vessel, Halacha, permutation test, within-woman permutation, multinomial sampling, natural family planning, Jewish law, polynomial hierarchy, dependence structure, multivariate randomization test

1. Introduction

Menstrual cycles exhibit substantial variability both within and across individuals [2, 10, 17]. The cycle length distribution is approximately bell-shaped with a pronounced right tail (skewness ≈ 1.3 in the present dataset), reflecting a small proportion of unusually long cycles. Within-person consistency is moderate: intraclass correlation coefficients for cycle length are typically in the range of 0.3–0.5 [3], indicating meaningful regularity beneath the surface variability. Bayesian hierarchical state-space models have been proposed to exploit this temporal structure for cycle-length prediction [1], confirming that within-woman dynamics carry genuine predictive information.

In the Jewish religious tradition, this regularity is also subject to detailed legal analysis. Halachic law (*Halacha*, Jewish religious law) requires that a woman anticipate the onset of her next menstruation based on patterns discernible in her prior cycle history. These anticipatory rules — the *halachic vessel rules* — govern a system of abstinence obligations central to the observance of *taharat hamishpacha* (family purity: the traditional Jewish system of laws governing marital intimacy in relation to the menstrual cycle). Of the 17 vessel pattern types documented in halachic literature, 10 require date or time-of-day data unavailable in standard cycle-length records; the remaining 7 are testable empirically. Two of these 7 (Dilug Chozer Chalila and Haflaga Chozer Chalila (Hebrew: repeating-cycle arithmetic progression and repeating fixed interval, respectively) occurred zero times in the present dataset and are not reported separately. The five patterns with positive observed frequencies are analysed here.

These five patterns form a *polynomial hierarchy*: each describes a *haflaga* (Hebrew: inter-menstrual interval, $H_k = L_k + 1$) sequence that follows a polynomial of a specific degree in cycle index. Haflaga, Week, and Week-Dilug are all degree-0 (constant) sequences; Dilug is degree-1 (linear); Dilug-in-Dilug is degree-2 (quadratic). This hierarchy generates algebraic structural relationships between the patterns — one of which (Haflaga-30 $\subseteq$ Week-Dilug) is a strict logical containment, and another (Haflaga–Dilug-in-Dilug proximity) an algebraic approximation rather than a logical implication — with implications for how their joint frequencies should be tested.

Despite the centuries-long practical application of vessel rules — and the modern availability of large longitudinal cycle datasets — no published study has formally tested whether

these patterns occur in empirical data at rates exceeding chance. This is a non-trivial question: if vessel rules merely detect random fluctuations, their anticipatory validity is weakened. Conversely, if they depart significantly from chance expectations, their empirical basis is independently corroborated.

The present study addresses this gap by applying two complementary null models — a permutation test [5] and a multinomial sampling null [16] — to a longitudinal NFP dataset [6]. We derive per-pattern p-values, effect sizes, and a joint multivariate test that accommodates the dependence structure among pattern counts without assuming independence. The joint test represents a methodological contribution [16], inspired by the simulation-based testing philosophy of King et al. [9] though distinct from their density-based approach; related ideas appear in multivariate permutation tests using quadratic forms [11]; a formal theoretical treatment is planned in future work. We also characterise the mathematical dependence structure through pairwise rate-based Spearman correlation tests and derive the dependencies analytically from the pattern definitions.

This study extends a line of quantitative halachic scholarship [13–15]. To our knowledge it is the first to test vessel pattern frequencies against an explicit randomisation null, though we note this claim has not been verified by a systematic literature search.

2. Background

2.1 The Polynomial Hierarchy of Vessel Rules

Let $L_1, L_2, \dots, L_n$ denote the menstrual cycle lengths (in days) of a single woman, and define the *haflaga* $H_k = L_k + 1$ (the interval from the first day of menstruation k to the first day of menstruation $k + 1$, following halachic convention [14]). The five vessel pattern types are:

Haflaga. $H_k = H_{k-1} = H_{k-2}$ (three or more consecutive equal haflaga values, any value).

Degree-0 polynomial (constant, unrestricted). Minimum run: 3 cycles.

Dilug. $H_k - H_{k-1} = H_{k-1} - H_{k-2} = d \neq 0$ (constant non-zero first difference). Degree-1 polynomial (linear). Minimum run: 3 cycles.

Dilug-in-Dilug. $(H_k - H_{k-1}) - (H_{k-1} - H_{k-2}) = (H_{k-1} - H_{k-2}) - (H_{k-2} - H_{k-3}) = d \neq 0$

(constant non-zero second difference). Degree-2 polynomial (quadratic). Minimum run: 4 cycles.

Week. $H_k = H_{k-1}$, $H_k \in \mathcal{W}$ where $\mathcal{W} = \{H_{\min} + 3, H_{\min} + 10, \dots\}$ (values spaced 7 apart, corresponding to the same weekday at each menstruation). Degree-0 polynomial anchored to weekly values. Minimum run: 2 cycles.

Week-Dilug. $H_k = H_{k-1} = 30$ (two or more consecutive haflaga values both equal to 30). The value $H = 30$ (29 actual days = 4 weeks +1 day) is the unique haflaga for which each successive menstruation falls one weekday later, producing an arithmetic progression of onset weekdays — hence the name *Week-Dilug* (Hebrew: *shavu'a be-dilug*, week with arithmetic progression; [16]). Week-Dilug thus combines the constant-interval property of Haflaga with the arithmetic-progression concept of Dilug, applied to the weekday domain rather than the cycle-length domain. Degree-0 polynomial (constant) at the canonical 30-day anchor. Minimum run: 2 cycles.

The hierarchy Haflaga (degree 0) $\rightarrow$ Dilug (degree 1) $\rightarrow$ Dilug-in-Dilug (degree 2) follows increasing polynomial order in cycle index. Week and Week-Dilug are specialisations of the degree-0 case anchored to calendar values.

2.2 Mathematical Dependence Structure

The polynomial hierarchy generates two analytically derivable dependence relationships between the remaining pattern pairs, which we verify empirically in Section 4.5:

Haflaga and Dilug-in-Dilug — proximity in the hierarchy. — When the second difference $|d|$ of a Dilug-in-Dilug sequence is small, its first differences are nearly equal and the haflaga sequence is nearly constant — i.e. nearly a Haflaga. In the data, 16 of 27 observed DiD sequences have $|d| = 1$, making them almost indistinguishable from a constant sequence. Both patterns further share the same structural prerequisite: a minimum run of ≥ 3 consecutive cycles satisfying a regularity condition, selecting the same women with high within-person cycle stability (within-person SD: Haflaga women = 1.54 days vs. no-vessel women = 3.17 days).

Dilug and Week-Dilug — logical incompatibility. — Dilug requires $\Delta H_k \neq 0$ (strictly non-constant differences); Week-Dilug requires $H_k = H_{k-1} = 30$ ($\Delta H = 0$). These conditions are mutually exclusive within the same cycle window, creating a negative dependence tendency.

2.3 Prior Work

Menstrual cycle statistics. — Variability and regularity of cycle length have been extensively characterised [2, 3, 10, 17]. Bortot et al. [1] developed a Bayesian hierarchical state-space model for sequential prediction of cycle lengths, exploiting within-woman temporal dependence; their model provides a benchmark for cycle-length predictability.

Quantitative halachic analysis. — Ross [15] modelled cycle transitions with Markov chains; [13] evaluated reliability of self-reported menstrual data in a halachic context; [14] applied statistical analysis to abstinence laws. The present study adds, to our knowledge, the first formal hypothesis test of whether vesset patterns occur at rates distinguishable from chance in real data (no systematic literature search was performed to confirm priority).

Randomization-based testing. — Permutation tests [5, 8] provide exact finite-sample p-values under exchangeability. The multinomial sampling null [16] draws each woman’s cycles independently from the fitted empirical distribution, directly encoding the null hypothesis of cycle independence. For the joint test, King et al. [9] proposed a general framework for testing a vector of statistics via simulation of the joint density, using a multivariate kernel density estimator to define a smallest-acceptance-region test. The present joint test differs: we use the Euclidean distance from the null mean vector as the test statistic, which is simpler, requires no density estimation, and has an intuitive geometric interpretation. A formal theoretical comparison and analysis of the distance-based approach is planned in future work [16].

3. Methods

3.1 Dataset

We analyzed a publicly available longitudinal dataset originally collected by Fehring et al. [6] in a randomised trial comparing two internet-supported natural family planning methods, archived at Marquette University [7]. After excluding cycles shorter than 18 days or longer than 54 days

[2], the final sample comprised **1,554 cycles from 118 women** ($M = 13.2$ cycles per woman, $SD = 6.1$, range 5–45).

Raw cycle lengths ranged from 18 to 54 days ($M = 29.3$, $SD = 3.9$, $Mdn = 28.5$, skewness = 1.29). The distribution is approximately bell-shaped with a right tail, consistent with the literature [2, 10]. Women ranged in age from 21 to 43 years ($M = 30.6$, $SD = 5.7$); parity data were not recorded in the archived dataset. The dataset originates from two charting cohorts ($n_0 = 92$ and $n_1 = 67$ women in the raw data prior to exclusions; $n_0 = 69$ and $n_1 = 49$ women in the filtered sample; cohort membership was not used as a sensitivity variable in the primary analysis, though it is available for future subgroup investigations). The minimum of 5 cycles per woman in the analysed dataset reflects the observed minimum after the cycle-length exclusion criteria were applied (18–54 days; complete records); no additional minimum-cycle-count exclusion was imposed. All cycle lengths were incremented by 1 to conform to the halachic haflaga convention ($H_k = L_k + 1$), yielding $M = 30.3$, $SD = 3.9$.

The within-person variability structure is relevant to the Haflaga result: women who establish a Haflaga vessel ($n = 20$) have a mean within-person cycle-length SD of 1.54 days ($SD_{\text{between}} = 0.59$), compared to 3.17 days ($SD_{\text{between}} = 1.75$) for women who establish no vessel at all ($n = 35$). A Mann–Whitney test confirms this difference is highly significant ($U = 133$, $p < .001$, two-sided; Welch $t = -5.05$, $df = 45.5$, $p < .001$; 95% CI for mean difference: $[-2.29, -0.98]$ days; Cohen’s $d = -1.13$). These results suggest that Haflaga is associated with lower observed cycle-length variability. Note that within-person SD estimates depend on the number of observed cycles per woman (dataset range: 5–45), so estimates for women with few recorded cycles carry greater sampling uncertainty. Figure 1 shows the cycle length distribution and per-woman variability. Figure 2 summarises the data flow from source dataset to analysed sample.

Figure 0. Menstrual cycle length distribution (filtered dataset)

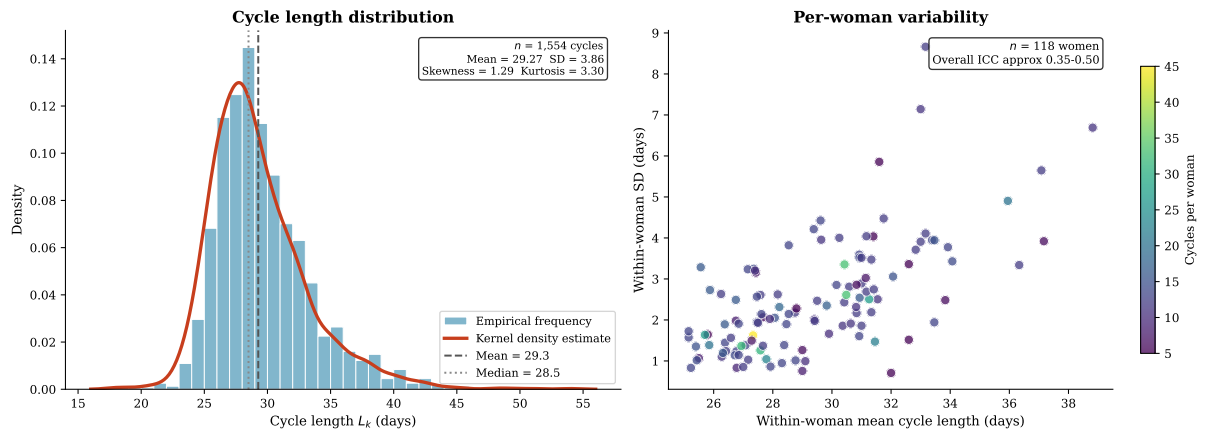


Figure 1: Cycle length distribution (filtered dataset). *Left:* histogram and kernel density estimate of raw cycle lengths L_k , with mean (dashed) and median (dotted) marked. The distribution is right-skewed (skewness = 1.29). *Right:* per-woman mean vs. within-person SD, shaded by cycle count. Women with more cycles tend to show lower variability, consistent with selection of naturally regular chartists.

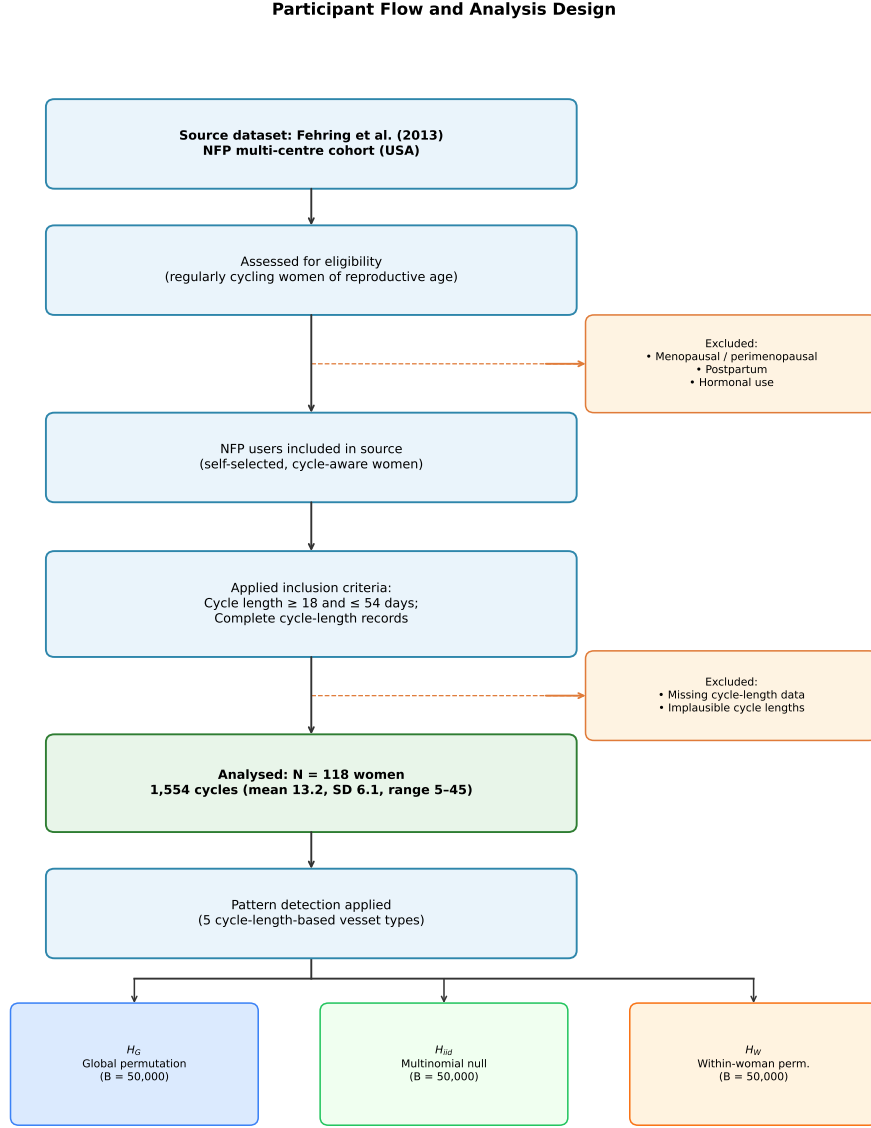


Figure 2: Participant and data flow diagram. The source dataset (Fehring et al., 2013) comprises data from a multi-centre NFP cohort. After applying inclusion criteria (cycle length 18–54 days; complete cycle-length records), the analysed dataset comprised 118 women contributing 1,554 cycles. Five pattern types were detected per woman. Three complementary null models were then applied: H_G (global permutation, $N = 50,000$ iterations), H_{id} (multinomial null, $N = 50,000$ iterations), and H_W (within-woman permutation null, $N = 50,000$ iterations per run).

3.2 Pattern Detection

Pattern counting was performed within each woman’s sequence independently; sequences were never merged across women. Patient boundary pairs (s_i, f_i) were encoded explicitly and all counting loops were constrained to $[s_i, f_i]$. Extended runs of the same pattern were

counted as a single occurrence to avoid double-counting (e.g., four consecutive equal haflaga values constitute one Haflaga, not two). If a qualifying run was interrupted by a discordant cycle and a new qualifying run of the same type began later in the same woman’s sequence, that resumed run was counted as a separate event; the counts therefore reflect all distinct qualifying runs across each woman’s full history, not only the initial fixation event. All five functions were validated against manually annotated examples and are available at <https://github.com/dvirross/VesetVectorTest>.

One counting implication follows directly from the definitions: Week-Dilug requires $H_k = H_{k-1} = 30$ (two consecutive haflaga values equal to 30), whereas Haflaga-30 requires three consecutive haflaga values equal to 30. Therefore every Haflaga-30 sequence automatically generates a Week-Dilug event in its first two cycles:

$$\{\text{Haflaga-30 sequence}\} \subseteq \{\text{Week-Dilug sequence}\}.$$

This is set-theoretic inclusion, not merely statistical correlation, and it means the Haflaga and Week-Dilug counts are not independent: the 3 Haflaga sequences with $H = 30$ each contribute directly to the Week-Dilug count of 18.

3.3 Null Models

Three null models address distinct scientific questions:

H_G (global exchangeability): All 1,554 cycle lengths exchangeable across women. Tests whether patterns exceed population-level chance.

H_W (within-woman exchangeability): Each woman’s cycles permuted uniformly at random within her own sequence (without replacement; observation indices are permuted, not unique values, so ties are handled correctly). Tests the null of *within-woman exchangeability of the observed sequence*: whether temporal ordering creates excess patterns beyond what each woman’s own cycle-length multiset predicts. This is not a general test of serial independence; a process with autocorrelation or latent trends could still fail to reject this null if those features do not produce additional exact veset events.

H_{iid} (multinomial i.i.d.): Each woman's cycles drawn i.i.d. from $\hat{F}$. Stronger than H_G : replaces observed values, not just ordering.

H_G and H_{iid} are the global nulls (permutation and multinomial respectively); H_W is the within-woman null. All three were run for $N = 50,000$ iterations (seed 17). Table 1 summarises the three null models side by side.

Table 1: Comparison of the three null models. Each model preserves some aspects of the data structure while randomising others, encoding a distinct scientific question.

	H_G (global perm.)	H_{iid} (multinomial)	H_W (within-woman)
Unit randomised	Individual cycle records, pooled across all women	Pattern occurrence per cycle, drawn i.i.d.	Cycle order within each woman's own sequence
What is preserved	Total cycle count; marginal haflaga distribution	Total count; observed marginal proportions	Each woman's multiset of cycle lengths and cycle count
What is destroyed	Woman identity; within-woman ordering	All dependence; temporal structure	Within-woman temporal ordering only
Independence assumed	None	Full i.i.d. (across time and women)	None (conditional permutation)
Scientific null	Patterns arise from random assignment of cycles to women	Patterns arise from independent random events	Patterns arise from random ordering of each woman's own cycles
Implementation	<code>np.random.permutation</code> on pooled haflaga values	Multinomial draw with observed proportions	Per-woman <code>np.random.permutation</code> (without replacement)

Permutation test. — At each iteration, the $n = 1,554$ haflaga values are randomly permuted without replacement. This preserves the exact empirical marginal distribution and is equivalent to assuming all $n!$ orderings are equally probable under H_0 [5].

Multinomial null. — At each iteration, for each woman i with n_i cycles, a fresh sequence $\tilde{H}_{i,1}, \dots, \tilde{H}_{i,n_i} \stackrel{\text{i.i.d.}}{\sim} \hat{F}$ is drawn, where $\hat{F}$ is the empirical CDF of all 1,554 observed haflaga values. This null directly encodes the hypothesis that cycles are *independent* across time within each woman [16]. It differs from a naive global bootstrap (resampling with replacement from the pooled distribution, which is similar in spirit to the multinomial null) in that each woman draws from the *same* marginal $\hat{F}$, ignoring between-woman heterogeneity in cycle-length distributions. The within-woman permutation null (H_W) addresses this by conditioning each woman’s simulated sequence on her *own* observed cycle-length multiset. A within-woman bootstrap (sampling with replacement from each woman’s own empirical distribution) was evaluated as an alternative but rejected: because consecutive i.i.d. draws are independent whereas permutation draws are negatively correlated (using a value reduces its availability at the next position), the bootstrap inflates the expected count of repetition-based patterns such as Haflaga — from a null mean of 31.9 under permutation to 54.5 under the bootstrap (a 71% increase verified by $B = 50,000$ simulations). Permutation is therefore preferred as the exact conditional randomisation test for the stated exchangeability null; the bootstrap corresponds to a different iid-within-woman null, and the two procedures are not directly comparable as more or less conservative in general — their different expected counts reflect the different scientific questions they encode.

3.4 Statistical Inference

For each pattern j and null model m we computed: the observed count O_j ; the null mean $\bar{S}_j^{(m)}$ and SD $\sigma_j^{(m)}$; the z-score $z_j^{(m)} = (O_j - \bar{S}_j^{(m)})/\sigma_j^{(m)}$; and the two-sided empirical p-value $p_j^{(m)} = 2 \min\left(\frac{b_j^+ + 1}{B + 1}, \frac{b_j^- + 1}{B + 1}\right)$, where $b_j^+ = \#\{S_{j,r}^{(m)} \geq O_j\}$, $b_j^- = \#\{S_{j,r}^{(m)} \leq O_j\}$, and $B = 50,000$, following the Monte Carlo convention of Davison & Hinkley [4]. All marginal tests use two-sided p-values; the direction and magnitude of departure from the null are separately reported via z-scores. The joint Mahalanobis distance is inherently two-sided — it measures displacement from the null centre in any direction — and serves as an omnibus test of whether the observed count vector is jointly unusual.

The **joint multivariate test** uses the **Mahalanobis distance** as the test statistic:

$$D_M^{(m)} = \sqrt{(\mathbf{O} - \boldsymbol{\mu}^{(m)})^\top (\hat{\Sigma}^{(m)})^{-1} (\mathbf{O} - \boldsymbol{\mu}^{(m)})}, \quad (1)$$

with p-value equal to the fraction of null replicates where the simulated D_M exceeded the observed value. Unlike Bonferroni–Holm correction — which, while valid under arbitrary dependence, is conservative when tests are positively correlated — this joint permutation test operates on the full count vector simultaneously, respecting the dependence structure. Note that this is a single global test of the multivariate null, not an FWER procedure across the five component tests. The joint Mahalanobis test is the **primary multivariate inferential procedure**; the individual Holm-corrected marginal tests are reported as **secondary descriptive summaries** of pattern-specific effect sizes and should not be used in place of the joint test for global inference. Conceptually related quadratic-form permutation tests exist (e.g., energy-distance statistics; 9) — the contribution here is an adaptation of this framework to dependent, sequentially-defined count vectors in longitudinal cycle data; to our knowledge this specific application is new. Mahalanobis distance correctly standardises each component by its null variance and accounts for inter-component correlations. Non-diagonal terms in $\hat{\Sigma}$ are expected a priori: the set-theoretic containment of Haflaga-30 within Week-Dilug and the polynomial proximity of Dilug-in-Dilug to Haflaga (Section 2.2) imply that the five pattern counts are not marginally independent, so a diagonal standardisation would misrepresent the dependence structure. $\hat{\Sigma}^{(m)}$ is estimated empirically from the $B = 50,000$ simulated count vectors under each null model: $\hat{\Sigma}^{(m)} = \frac{1}{B-1} \sum_{r=1}^B (\mathbf{S}^{(m,r)} - \bar{\mathbf{S}}^{(m)})(\mathbf{S}^{(m,r)} - \bar{\mathbf{S}}^{(m)})^\top$, where $\mathbf{S}^{(m,r)}$ is the count vector from replicate r and $\bar{\mathbf{S}}^{(m)}$ is the Monte Carlo mean; this is the sample covariance of the null replicates, not the sample covariance of the observations. The result is a 5×5 matrix. Condition numbers of 4.4 (permutation null) and 5.4 (multinomial null) confirm that the matrix is well-conditioned and the inversion is numerically stable; note that a low condition number establishes numerical stability of the inversion, not that Mahalanobis distance is the most powerful statistic for every scientific alternative (the statistic detects significant departure in any direction, not specifically the positive direction; the direction of displacement is separately reported via the individual z -scores). The Euclidean displacement $D_E^{(m)} = \|\mathbf{O} - \boldsymbol{\mu}^{(m)}\|_2$ is

shown geometrically in the figures as the length of the dashed displacement line.

The **primary joint test** is the focused Mahalanobis test restricted to the three structurally dependent patterns — Haflaga, Week-Dilug, and Dilug-in-Dilug. This choice is justified on purely structural grounds, established before inspecting the data:

- **Structural basis for inclusion.** Haflaga-30 $\subseteq$ Week-Dilug by set-theoretic containment (Section 2.2), creating a deterministic co-occurrence dependency. Dilug-in-Dilug is the degree-2 polynomial neighbour of degree-0 Haflaga in the hierarchy, creating an algebraic proximity. These three patterns therefore form a natural analytical unit with prior structural predictions for their joint behaviour.
- **Structural basis for exclusion.** Dilug (degree-1) and Week (degree-0 anchored to a calendar grid) have no structural containment or polynomial proximity relationship with the three included patterns. Dilug yields a non-significant marginal result (Holm adj $p = .346$) with no analytical prediction for its relationship to the primary triad; including it would conflate structurally motivated testing with a pattern that is only hypothesis-generating at this stage.

Null and alternative. The 3-pattern null hypothesis $H_0^{(3)}$ states that the observed count vector $(O_{\text{Haf}}, O_{\text{WD}}, O_{\text{DiD}})$ follows the joint null distribution of the corresponding 3-pattern subvector under H_W (or H_G for the global variant) — that is, $D_M^{(3)} \leq D_M^{(3,r)}$ for the fraction of null replicates that determines the p -value. The alternative $H_1^{(3)}$ is that the observed vector is displaced from the null centre in some direction — at least one count exceeds its null expectation once inter-pattern correlations are accounted for via $\hat{\Sigma}^{(3)}$ (the 3×3 submatrix of the full null covariance). The statistic is sensitive to displacement in any direction, with Haflaga receiving greater weight due to its smaller null variance. The Mahalanobis test is particularly apt here because the three patterns are non-independent by construction (Section 2.2), and the test accounts for this dependence structure through $\hat{\Sigma}^{(3)}$ rather than requiring independence.

The 3-pattern Mahalanobis statistic applies Eq. 1 to the subvector $(O_{\text{Haf}}, O_{\text{WD}}, O_{\text{DiD}})$ with a 3×3 covariance submatrix (condition number 2.6; well-conditioned). The **5-pattern test** over all patterns is reported as a robustness check.

Sensitivity. — The joint test is sensitive primarily to the Haflaga component, which contributes the largest standardised excess ($z = 4.51$) and — after variance standardisation by the Mahalanobis metric — the largest weight. Under the global permutation null, the observed $D_M = 5.10$ exceeds the 99.9th percentile of the null distribution.

Pairwise **Spearman rank correlations** were computed between per-woman pattern rates (number of pattern events divided by number of cycles). Using rates rather than binary indicators mitigates the cycle-count exposure confound: women with more cycles have more opportunities to satisfy any pattern, and normalising by cycle count reduces but does not eliminate this exposure difference (the permutation randomises the rate vector rather than conditioning on cycle count). Two-sided permutation p -values (50,000 permutations of one variable’s rate vector) with Holm–Bonferroni correction across the 10 pairs were used for inference. $\alpha = .05$ throughout.

Simulation pipeline. — Algorithm 1 provides a formal pseudocode description of the full simulation pipeline for reproducibility.

Minimum run-length examples. — To clarify the counting convention, consider a single woman’s haflaga sequence. A run of length 2: (29, 29) qualifies as Week-Dilug (if $H = 30$) or Week but *not* as Haflaga (which requires at least 3). A run of length 3: (28, 28, 28) qualifies as Haflaga (and as Week if $28 \in \mathcal{W}$). A run of length 4: (27, 29, 31, 33) qualifies as Dilug ($d = +2$, constant first difference, ≥ 3 cycles) but also as Dilug-in-Dilug only if a second constant non-zero difference exists — this sequence has second differences $(2 - 2, 2 - 2) = (0, 0)$, so it does *not* qualify as Dilug-in-Dilug (which requires $d \neq 0$ in the second differences). Instead (27, 28, 30, 33) has first differences (1, 2, 3) and second differences (1, 1), giving $d = 1 \neq 0$, so it qualifies as Dilug-in-Dilug (minimum 4 cycles).

Algorithm 1 Full simulation pipeline

Require: Dataset $\mathcal{D} = \{(H_{i,1}, \dots, H_{i,n_i})\}_{i=1}^N$ (N women), patterns $j = 1, \dots, 5$, iterations B , seed

- 1: Compute observed counts $\mathbf{O} = (O_1, \dots, O_5)$ via pattern detection
- 2: **for** $r = 1$ to B **do**
- 3: // Global permutation null (H_G): shuffle all $\sum_i n_i$ haflaga values
- 4: $\mathbf{H}^{(G)} \leftarrow \text{Permute}(\bigcup_i \{H_{i,k}\})$; re-assign to women preserving n_i
- 5: $\mathbf{S}^{(G,r)} \leftarrow \text{Count}(\mathbf{H}^{(G)})$
- 6: // Multinomial null (H_{iid}): draw each $H_{i,k}$ iid from marginal distribution
- 7: $\mathbf{H}^{(M)} \leftarrow \text{Multinomial}(p_1, \dots, p_K; \text{total } \sum_i n_i)$
- 8: $\mathbf{S}^{(M,r)} \leftarrow \text{Count}(\mathbf{H}^{(M)})$
- 9: // Within-woman permutation null (H_W): shuffle within each woman
- 10: **for** $i = 1$ to N **do**
- 11: $\mathbf{H}_i^{(W)} \leftarrow \text{Permute}(H_{i,1}, \dots, H_{i,n_i})$ ▷ without replacement
- 12: **end for**
- 13: $\mathbf{S}^{(W,r)} \leftarrow \text{Count}(\mathbf{H}^{(W)})$
- 14: **end for**
- 15: Estimate $\hat{\Sigma}^{(m)} \leftarrow \text{SampleCov}(\{\mathbf{S}^{(m,r)}\}_{r=1}^B)$ for each $m \in \{G, M, W\}$
- 16: **for** $m \in \{G, M, W\}$ **do**
- 17: Null mean $\boldsymbol{\mu}^{(m)} \leftarrow B^{-1} \sum_r \mathbf{S}^{(m,r)}$
- 18: Mahalanobis statistic: $D_M^{(m)} \leftarrow \sqrt{(\mathbf{O} - \boldsymbol{\mu}^{(m)})^\top (\hat{\Sigma}^{(m)})^{-1} (\mathbf{O} - \boldsymbol{\mu}^{(m)})}$
- 19: $p_j^{(m)} \leftarrow (|\{r : S_j^{(m,r)} \geq O_j\}| + 1)/(B + 1)$ for each pattern j ▷ two-sided
- 20: $p_{DM}^{(m)} \leftarrow (|\{r : D_M^{(m,r)} \geq D_M^{(m)}\}| + 1)/(B + 1)$
- 21: **end for**

4. Results

4.1 Per-Pattern Hypothesis Tests

Observed counts were: Haflaga = 26, Dilug = 61, Week = 32, Week-Dilug = 18, Dilug-in-Dilug = 27. Of the 26 Haflaga sequences, 3 had value $H = 30$, each automatically constituting

a Week-Dilug event (set-theoretic inclusion); of the 27 Dilug-in-Dilug sequences, 16 had second difference $|d| = 1$ (near-Haflaga). Table 2 presents the full results under both null models. Figures 3 and 4 display the null distributions and effect sizes.

Table 2: Permutation and multinomial null-model results ($N = 50,000$ iterations). *Null Mean* and *Null SD* are from the permutation null. Bold p -values are significant at $\alpha = .05$ (two-sided).

Pattern	Obs.	Null	Null SD	z	Perm. p	Multi. p
Mean						
Haflaga	26	11.30	3.26	+4.51	< .001	< .001
Dilug	61	48.93	6.60	+1.83	.087	.086
Week	32	27.00	4.03	+1.24	.133	.188
Week-Dilug	18	16.27	3.28	+0.53	.345	.366
Dilug-in-Dilug	27	27.17	5.04	−0.03	.543	.544

Figure 1: Empirical null distributions (PMF) for each vessel pattern

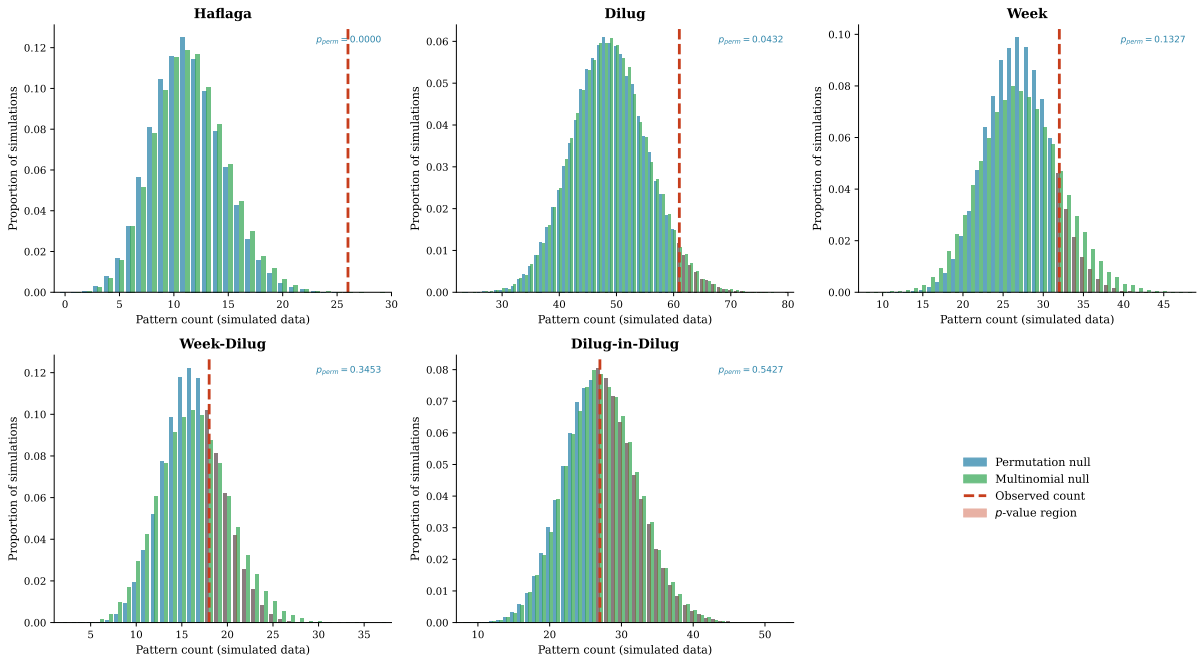


Figure 3: Empirical null distributions (shown as smoothed density estimates) vs. observed counts for each vessel pattern under the permutation (solid curve) and multinomial (dashed curve) null models. The underlying null distributions are discrete; the smoothed curves are shown for visual comparison only, with the primary inference based on the empirical PMF (fraction of $B = 50,000$ simulations meeting or exceeding the observed count). The vertical dotted line marks the observed count; the shaded region is the two-sided p -value area. Haflaga shows a clearly right-displaced observed value; Dilug-in-Dilug falls on its null mean.

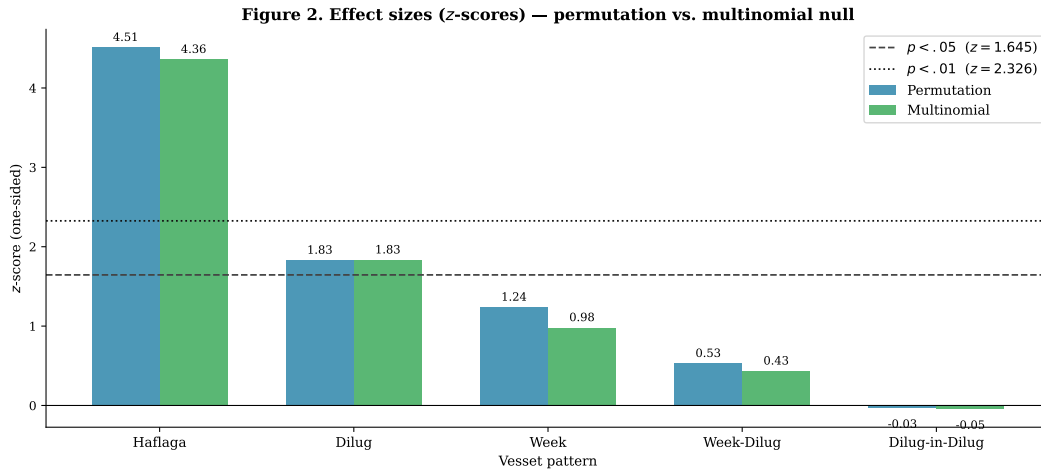


Figure 4: Effect sizes (z -scores) under permutation (solid bars) and multinomial (hatched bars) null models. Dashed and dotted lines mark $\alpha = .05$ ($z = 1.645$) and $\alpha = .01$ ($z = 2.326$) thresholds. z -scores differ between nulls (e.g., Week: 1.24 vs. 0.98) because each null has its own mean and SD. Only Haflaga and Dilug exceed the $\alpha = .05$ threshold; both null models are in close agreement.

Haflaga showed the strongest effect: 26 occurrences vs. a null mean of 11.30, more than 4.5 standard deviations above chance ($z = 4.51$, exact Monte Carlo $p = 3/50,001 \approx .00006$). Crucially, under the permutation null only **2 of 50,000** simulations reached or exceeded 26 — the observed count ties the simulation maximum. Applying Holm–Bonferroni correction across the five pattern tests, only Haflaga remains significant after FWER control (Holm-adjusted $p = .0003$). Dilug is not significant ($p = .087$, Holm-adjusted $p = .346$) and should be regarded as an exploratory finding requiring replication. Week, Week-Dilug, and Dilug-in-Dilug are not significant. These marginal results are complementary summaries; the primary multivariate inference is the joint Mahalanobis test (Section 4.4), which operates on the full count vector simultaneously without multiple-comparison correction.

4.2 Agreement Between Null Models

The permutation and multinomial null models are in close agreement across all five patterns (maximum difference in p -values: 0.055). Figure 5 shows the empirical CDFs, confirming the two nulls produce nearly identical distributions for each pattern. The close agreement is a numerical observation; whether and under what conditions the two procedures converge asymptotically depends on the specific sampling model and is not formally established here.

Figure 4: Empirical CDFs of null distributions for each vessel pattern

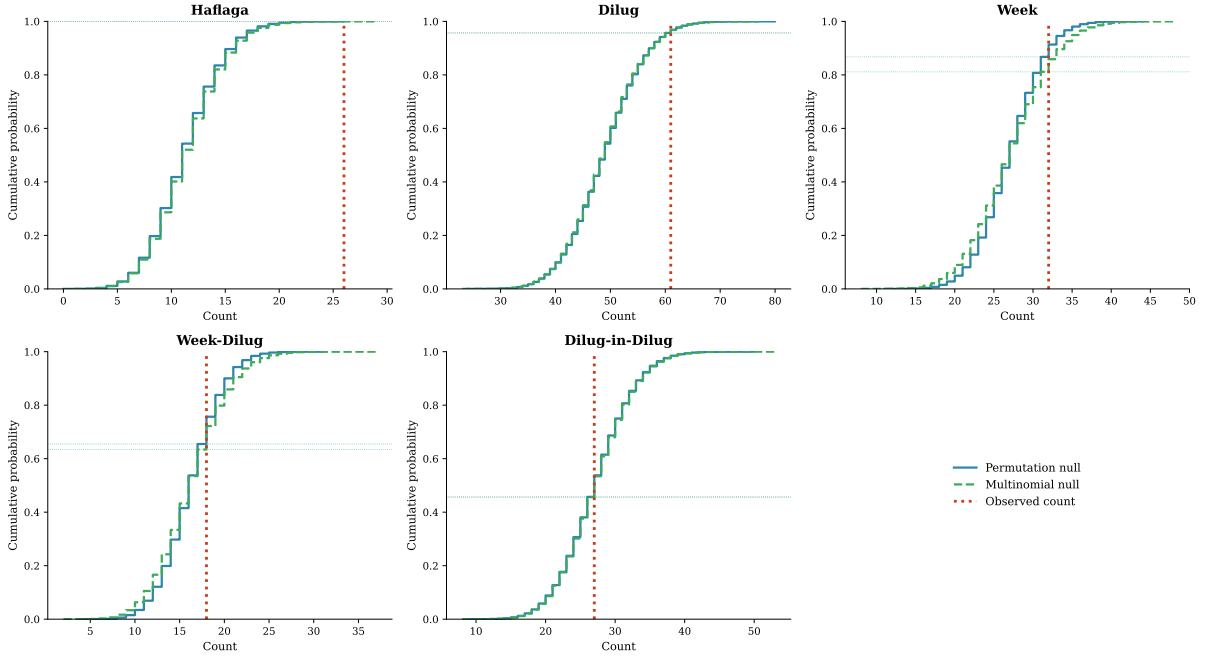


Figure 5: Empirical CDFs of null distributions for each pattern under permutation (solid curve) and multinomial (dashed curve) nulls. The vertical dotted line marks the observed count; horizontal dotted lines indicate $1 - p$. The close overlap confirms robustness to the choice of null model.

4.3 Within-Woman Permutation Analysis

The global permutation and multinomial null models permute or resample haflaga values *across* all 118 women, thereby removing between-woman heterogeneity in cycle regularity. As noted in Section 3.3, this means both global nulls test whether vessel patterns are more frequent than expected if all women drew from the same population distribution — not whether patterns are more frequent than expected *given each woman’s own level of regularity*. These are distinct scientific questions.

To address the within-woman question, we implemented a **within-woman permutation** null (H_W): for each woman independently, her cycle-length sequence is randomly permuted (shuffled without replacement), preserving her individual cycle-length *multiset* while randomising the temporal ordering. This null encodes the hypothesis that the *temporal ordering* of cycles within a woman does not create patterns beyond what her own cycle-length values (taken in random order) would predict. The null is specifically a conditional exchangeability null: it does not test general serial independence, and a process with autocorrelation or trend structure

could still fail to reject it if such structure does not translate into additional exact vessel-pattern occurrences.

Implementation. — For each of the 118 women independently, her cycle-length sequence is permuted uniformly at random (without replacement). All women are included with no exclusion threshold. The aggregate null is generated by summing the permuted count vectors across all 118 women, producing $B = 50,000$ realisations of the joint count vector.

Results. — Table 3 presents the results. Under the within-woman null, *no pattern is individually significant*. All five observed counts fall *below* their within-woman null means (all $z < 0$). Dilug-in-Dilug has the most extreme z-score ($z = -2.07$, two-sided $p = .040$, Holm-corrected $p = .198$); this does not survive multiple-comparison control and should be regarded as exploratory. The negative direction — the observed count (27) is well below the within-woman null mean (39.1) — indicates that real temporal ordering of cycles produces fewer DiD events than random shuffling of the same cycle lengths would predict, which is consistent with DiD detecting genuine second-order cycle regularity rather than chance co-occurrence of lengths. The joint test is also non-significant: Mahalanobis $D_M = 2.95$ ($p = .121$). Tables 3 and 5 both use the primary 50,000-run simulation (seed 17) to ensure full numerical consistency of μ_W , σ_W , and z_{obs} across all results.

Table 3: Within-woman permutation null (H_W) results ($N = 50,000$; all $n = 118$ women; each woman’s sequence permuted uniformly at random without replacement). All observed counts fall below their within-woman null means. p -values are two-sided: $2 \min((b^+ + 1)/(B + 1), (b^- + 1)/(B + 1))$.

Pattern	Obs.	Null Mean	Null SD	z	p
Haflaga	26	31.85	4.68	−1.25	.249
Dilug	61	67.07	7.36	−0.83	.455
Week	32	33.81	3.88	−0.47	.741
Week-Dilug	18	20.94	3.22	−0.91	.454
Dilug-in-Dilug	27	39.14	5.86	−2.07	.040
Joint: $D_M = 2.95$ ($p = .121$)					

Figure 6 illustrates the contrast between the global and within-woman nulls for each pattern. The shift in the null distribution is striking: for Haflaga, the global null is centred around 10, while the within-woman null is centred around 27 — far above the observed count of 26 (all 118 women).

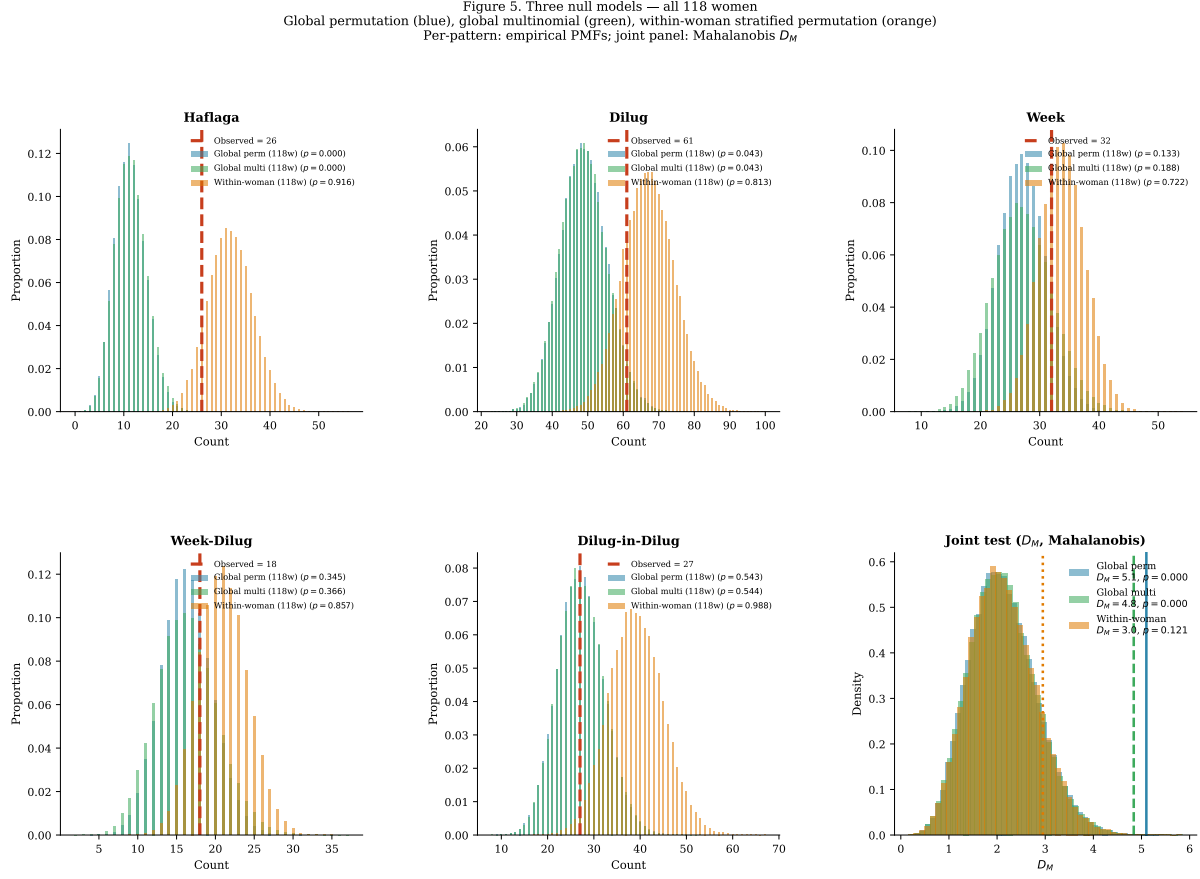


Figure 6: Null distributions (H_G , H_{iid} , H_W) for each pattern under three null models: global permutation (light fill), global multinomial (medium fill), and within-woman permutation (dark fill), all $N = 118$ women. Dashed vertical line: observed count. The joint panel shows Mahalanobis D_M histograms; both global nulls are highly significant ($p < .001$) while the within-woman null is non-significant ($D_M = 2.95$, $p = .121$).

Interpretation. — The two nulls answer different scientific questions and are both valid:

- The **global null** tests: *are vessel patterns more frequent than expected if all cycle lengths were exchangeable across all women?* The answer is yes for Haflaga ($p < .001$ after Holm correction), reflecting that vessel patterns are concentrated among women who are intrinsically more regular.
- The **within-woman null** tests: *given each woman's own level of cycle regularity, are*

the temporal arrangements unusually structured? The answer is no — under the within-woman exchangeability null, this dataset provides no detectable evidence that temporal ordering adds structure beyond each woman’s own cycle-length distribution.

Together, these results suggest that halachic vessel rules — particularly Haflaga — constitute an empirical codification of individual menstrual regularity: the rules correctly identify the subset of women whose cycles are intrinsically stable, and formalise that stability as an anticipatory pattern. Whether this reflects deliberate halachic wisdom or fortuitous alignment with biology is beyond the scope of the present analysis. A companion study examining the practical forecasting efficiency of halachic vessel rules — how accurately they predict the next menstruation onset, and at what cost in unnecessary abstinence days — is in preparation [12].

4.4 Joint Multivariate Test

Primary test: focused 3-pattern Mahalanobis. — The primary joint test operates on the structurally dependent subvector (Haflaga, Week-Dilug, Dilug-in-Dilug), justified on analytical grounds prior to data inspection (Section 3.4). The 3-pattern Mahalanobis statistic was $D_M = 4.52$ ($p < .001$) under the permutation null and $D_M = 4.36$ ($p < .001$) under the multinomial null. Figure 9 illustrates the displacement of the observed 3-pattern vector from the respective null clouds under all three null models, using these three patterns as the coordinate axes.

Robustness: 5-pattern Mahalanobis. — Extending the joint test to all five patterns yields $D_M = 5.10$ ($p < .001$, permutation) and $D_M = 4.84$ ($p < .001$, multinomial). Figure 7 shows the 5-pattern Mahalanobis null distributions for all three null models. The agreement across all variants confirms that the primary result is robust to the choice of pattern subset and null model.

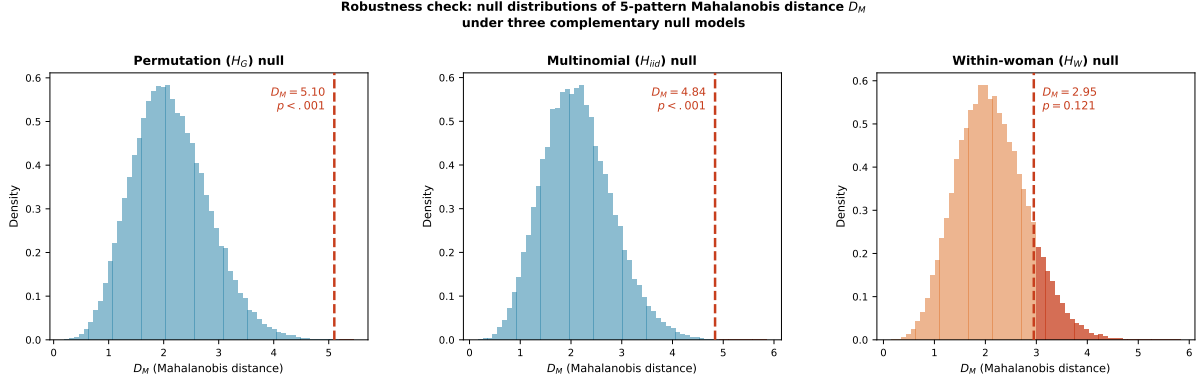


Figure 7: Robustness check: null distributions of the 5-pattern Mahalanobis distance D_M (Eq. 1) under the permutation (H_G , left), multinomial (H_{iid} , centre), and within-woman (H_W , right) nulls. The dashed line marks the observed D_M ; the shaded region is the upper-tail area corresponding to the empirical p-value. Under H_G and H_{iid} the observed vector lies far in the tail ($p < .001$); under H_W the displacement is non-significant ($D_M = 2.95$, $p = .121$).

4.5 Pairwise Dependence Tests and Focused Joint Test

The structural relationships of Section 2.2 motivate a prior prediction for the pairwise dependence tests, but the two predicted pairs have different logical status and should be interpreted accordingly.

Layer 1 — Deterministic logical structure. Haflaga-30 is a subset of Week-Dilug by definition: any woman who establishes a Haflaga sequence at $H = 30$ necessarily satisfies the Week-Dilug criterion. This is a logical implication of the pattern definitions, not an empirical finding.

Layer 2 — Exploratory empirical associations. Haflaga $\times$ Dilug-in-Dilug rests on the algebraic proximity argument (Section above): a DiD sequence with $|d| = 1$ is close to a constant, so co-occurrence with Haflaga is plausible but not logically entailed. This association, and all remaining pairs, are treated as exploratory. No positive association is analytically predicted for the remaining eight pairs.

These pairwise tests therefore serve as a **consistency check** against prior structural expectations: the question is not whether we can discover new associations, but whether the data are consistent with relationships derivable from the pattern definitions. All findings are exploratory and subject to the cycle-count exposure confound noted in Section 5.7.

Each pair is assessed via Spearman rank correlation between per-woman pattern rates, with two-sided permutation p -values (50,000 permutations) and Holm–Bonferroni correction

across all 10 pairs (see Section 3 for the rationale for using rates). Table 4 presents the results.

The data are compatible with the analytical predictions: both structurally predicted pairs show positive correlation. It is important to distinguish what these results establish. The set-theoretic containment ($\text{Haflaga-30} \subseteq \text{Week-Dilug}$) is a deterministic logical consequence of the pattern definitions — it does not require empirical verification, and a positive association here merely confirms the counting arithmetic, not an additional stochastic mechanism. The algebraic proximity between Haflaga and Dilug-in-Dilug, by contrast, is a probabilistic plausibility argument: the observed association is *compatible* with the proposed proximity mechanism, but a formal prediction requires an explicit stochastic model for the data-generating process, which is beyond the scope of this paper.

After Holm correction, $\text{Haflaga} \times \text{Dilug-in-Dilug}$ survives ($r_s = 0.27$, $p_{\text{perm}} = .003$, $\text{adj } p = .030$). $\text{Haflaga} \times \text{Week-Dilug}$ shows a positive association ($r_s = 0.23$, $p_{\text{perm}} = .010$) consistent with the structural prediction, but does not survive Holm correction ($\text{adj } p = .090$). All eight remaining pairs are non-significant (all $\text{adj } p \geq .206$), consistent with the absence of analytical predictions for those pairs. Figure 8 visualises the full correlation structure.¹

¹A robustness check using a binary co-occurrence analysis — coding each woman 1/0 for whether she ever established each pattern, and testing all 10 pairs with Fisher’s exact test plus permutation p -values — yielded the same two lowest p -values ($\text{Haflaga} \times \text{Dilug-in-Dilug}$: $p_{\text{perm}} = .004$, $\text{Holm adj } p = .040$; $\text{Haflaga} \times \text{Week-Dilug}$: $p_{\text{perm}} = .013$, $\text{Holm adj } p = .117$) and the same pattern of non-significance for all remaining pairs, confirming that the rate-based results are not sensitive to the choice of metric.

Table 4: Pairwise Spearman rate-based correlation tests between vessel types ($N = 118$ women). Each woman's pattern rate = event count / number of cycles. Pairs are sorted by p_{perm} . r_s : Spearman rank correlation; p_{perm} : two-sided permutation p -value (50,000 permutations); p_{Holm} : Holm–Bonferroni adjusted p -value across all 10 pairs. Direction: + = positive correlation; – = negative correlation. †: analytically predicted positive pair (Section 2.2). Bold: survives Holm correction ($p_{\text{Holm}} < .05$).

Pair	r_s	p_{perm}	p_{Holm}	Dir.
Haflaga $\times$ Dilug-in-Dilug†	0.27	.003	.030	+
Haflaga $\times$ Week-Dilug†	0.23	.010	.090	+
Dilug $\times$ Week-Dilug	–0.12	.206	1.000	–
Haflaga $\times$ Week	0.10	.272	1.000	+
Dilug $\times$ Week	0.09	.325	1.000	+
Week $\times$ Dilug-in-Dilug	0.09	.326	1.000	+
Week $\times$ Week-Dilug	0.07	.474	1.000	+
Week-Dilug $\times$ Dilug-in-Dilug	0.07	.483	1.000	+
Haflaga $\times$ Dilug	0.04	.676	1.000	+
Dilug $\times$ Dilug-in-Dilug	–0.02	.815	1.000	–

Pairwise Spearman correlations (rate-based)
 † analytically predicted * Holm adj $p < .05$

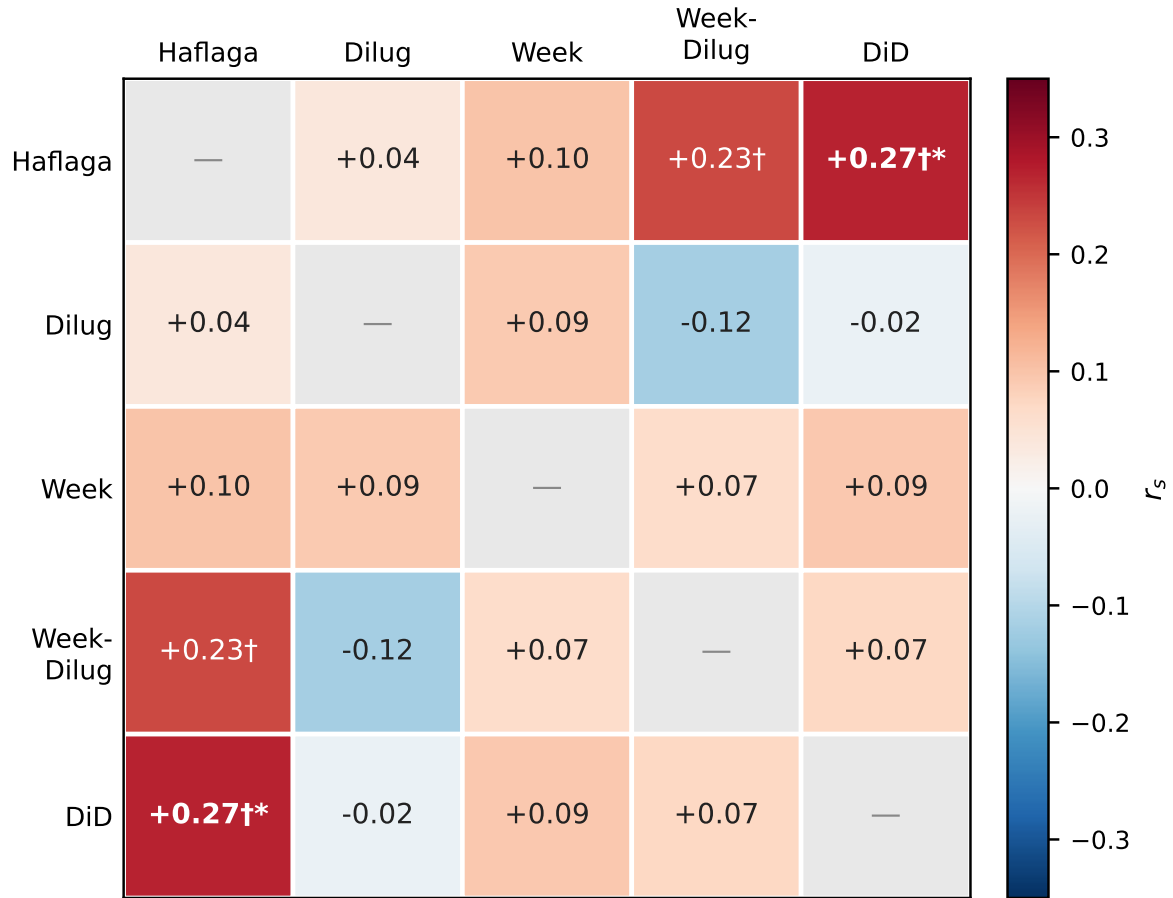


Figure 8: Pairwise Spearman correlation structure between vessel pattern types. Each cell shows r_s (rate-based Spearman correlation); darker shading indicates stronger positive association; lighter shading indicates weaker or negative association. Both analytically predicted pairs (Haflaga $\times$ Dilug-in-Dilug and Haflaga $\times$ Week-Dilug, marked †) show positive correlation; Haflaga $\times$ Dilug-in-Dilug survives Holm correction ($r_s = 0.27$, adj $p = .030$), compatible with the polynomial-hierarchy proximity argument of Section 2.2 (all pairwise tests are exploratory).

The primary 3-pattern Mahalanobis joint test is reported in Section 4.4. Figure 9 provides a geometric illustration of the displacement of the observed count vector in the (Haflaga, Week-Dilug, Dilug-in-Dilug) subspace under all three null models. The three patterns serve as coordinate axes; the remaining two patterns (Dilug, Week) are encoded by marker size and point shading.

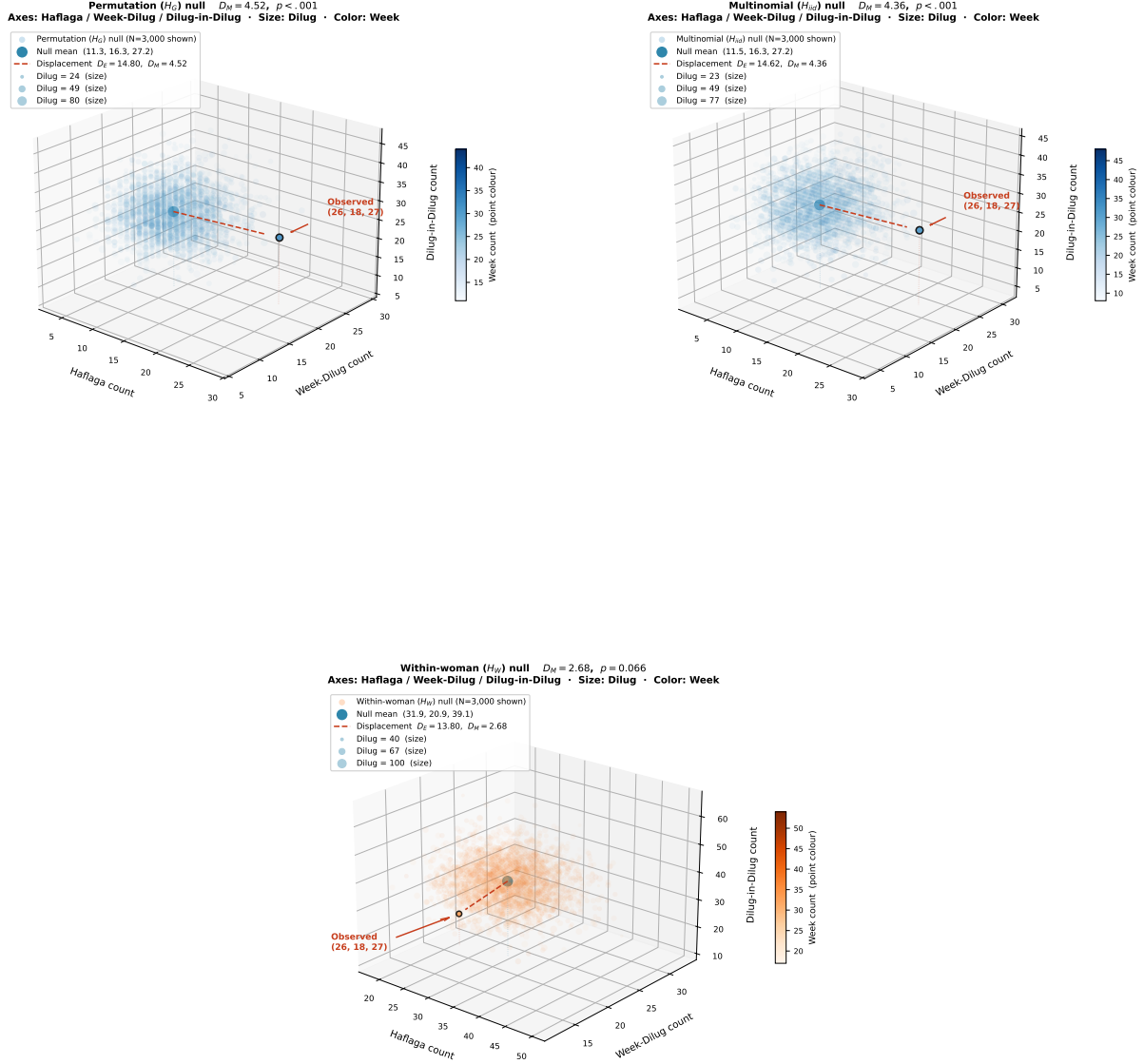
Focused 3-pattern joint test: null clouds vs. observed vector
under three complementary null models


Figure 9: Geometric illustration of the focused 3-pattern joint test under the **permutation** (H_G , top left), **multinomial** (H_{iid} , top right), and **within-woman** (H_W , bottom centre) nulls. Axes: Haftaga, Week-Dilug, Dilug-in-Dilug — the three structurally dependent patterns. Marker size encodes Dilug count; marker shading encodes Week count. The filled circle marks the null mean; the dashed line and arrow identify the observed vector (26, 18, 27). Under H_G and H_{iid} the observed vector lies at the periphery of the null cloud, displaced primarily along the Haftaga axis ($D_M = 4.52$ and 4.36 , both $p < .001$). Under H_W the observed vector falls within the null cloud and the displacement is non-significant (3-pattern $D_M = 2.68, p = .066$), consistent with the primary result (H_W 5-pattern: $D_M = 2.95, p = .121$; Table 3).

5. Discussion

This study provides the first formal statistical test of whether halachic vessel patterns occur at rates distinguishable from chance in empirical menstrual cycle data. The central result is

from the within-woman permutation null (H_W): no pattern is individually significant after Holm correction (all Holm-corrected $p \geq .198$; joint $D_M = 2.95$, $p = .121$), and all observed counts fall below their within-woman null means. We found no evidence that the temporal ordering of cycles produces excess vessel-pattern occurrences beyond those expected under within-woman exchangeability, conditional on each woman’s observed cycle-length multiset. This is a non-detection of excess under the exchangeability null, not a demonstration that serial dependence is absent: a process with autocorrelation, latent states, or gradual trends could still fail to reject this particular permutation null if such structure does not translate into additional exact vessel events.

Summary of confirmatory vs exploratory results. — For clarity, we distinguish three categories. **Pre-specified confirmatory:** (i) Haflaga overrepresentation under the global null ($p < .001$ after Holm correction); (ii) the non-significant H_W result (all Holm-corrected $p \geq .198$; joint $D_M = 2.95$, $p = .121$); (iii) the primary 3-pattern Mahalanobis joint test under the global null ($D_M = 4.52$, $p < .001$). **Exploratory:** Dilug ($p = .087$, Holm adj $p = .346$; hypothesis-generating only), pairwise dependence tests, and the focused joint test (motivated structurally but secondary). **Robustness checks:** 5-pattern joint test and multinomial null, confirming the global results are not sensitive to null model or pattern-subset choice.

Under the global permutation null, which pools cycles across all 118 women, *Haflaga* is strongly overrepresented ($z = 4.51$, $p < .001$ after Holm correction) and *Dilug* is not significant ($p = .087$, Holm adj $p = .346$). The joint Mahalanobis test under the global null confirms significant displacement ($D_M = 5.10$, $p < .001$), driven primarily by *Haflaga*. The global overrepresentation reflects concentration of patterns among women who are intrinsically more regular — a population-level heterogeneity effect, not evidence of within-woman temporal structure.

5.1 Pattern-Specific Findings

The *Haflaga* finding is the most striking result. With only 2 of 50,000 permutation simulations reaching or exceeding the observed count of 26 (the simulation ceiling), the result is highly significant under the global null and consistent with the expectation that *Haflaga* would be

concentrated among women with lower within-person cycle-length variability in this single non-Jewish NFP dataset. Women who establish Haflaga have within-person SDs of 1.54 days ($n = 20$) vs. 3.17 days for women who establish no vessel at all ($n = 35$; Mann–Whitney $p < .001$, Cohen’s $d = -1.13$; see Section 3.1). Forecasting performance and generalisability to other populations remain to be evaluated in future work.

The Dilug finding ($p = .087$ under the global null) does not approach significance and does not survive Holm–Bonferroni correction (adjusted $p = .346$); it should be treated as hypothesis-generating, requiring pre-registered replication in larger samples. The direction is consistent with arithmetic progressions arising from gradual biological trends (e.g., hormonal change, ageing, or weight change), but these explanations are speculative and await independent confirmation.

5.2 Mathematical Dependence Structure as a Unifying Observation

A distinctive feature of this study is that two of the structural relationships between pattern types derive from the pattern definitions themselves (Section 2.2). These should be distinguished carefully: one is a strict logical necessity, the other an algebraic approximation. To our knowledge, this distinction has not previously been stated explicitly in the halachic or statistical literature.

The three key relationships are: (i) **deterministic containment** — every Haflaga-30 sequence is also a Week-Dilug event by set-theoretic inclusion, so 3 of the 26 Haflaga cases count deterministically towards Week-Dilug; (ii) **algebraic proximity** — a Dilug-in-Dilug sequence with small second difference $|d| = 1$ deviates from a constant sequence by at most a few days, placing it behaviourally close to Haflaga; this is an algebraic approximation, not a logical implication (the bound depends on both $|b|$ and $|d|$, so a large first difference can produce substantial variation even with $|d| = 1$); 16 of the 27 observed DiD sequences have $|d| = 1$, and the positive association ($\chi^2 = 9.99$, $p_{\text{perm}} = .004$) is consistent with but not entailed by this proximity; (iii) **mutual exclusion at the window level** — a cycle pair cannot simultaneously satisfy Dilug ($\Delta^1 H \neq 0$) and Week-Dilug ($\Delta^1 H = 0$), producing the observed negative direction for that pair ($p = .300$, non-significant at the woman level).

The five pattern counts are therefore non-independent by construction. The Holm–Bonferroni and joint Mahalanobis procedures answer different inferential questions (FWER

control vs. a single global null), and the focused 3-pattern Mahalanobis test accounts for the inter-pattern covariance structure through $\hat{\Sigma}^{(3)}$.

5.3 The Joint Multivariate Test

The joint test (Eq. 1) standardises deviations by the full null covariance $\hat{\Sigma}$, accounting for inter-pattern correlations (condition numbers 4.4–5.4; all pairwise $|r| < 0.16$; see Section 3 for full details). Haflaga has the smallest null SD (3.26 vs. 6.60 for Dilug), so its deviation $(26 - 11.30) = 14.7$ units yields the largest standardised contribution.

The primary result — the focused 3-pattern Mahalanobis test — yields $D_M = 4.52$ ($p < .001$; 1 of 50,000 null replicates exceeded the observed D_M) under the permutation null and $D_M = 4.36$ ($p < .001$; 0 of 50,000 exceeded) under the multinomial null. The 5-pattern robustness check yields $D_M = 5.10$ ($p < .001$, permutation; 0 of 50,000 exceeded) and $D_M = 4.84$ ($p < .001$, multinomial; 0 of 50,000 exceeded). All four variants yield highly significant results, confirming that the primary finding is robust to the choice of pattern subset. Haflaga receives appropriately up-weighted contribution via the Mahalanobis metric, reflecting its larger standardised excess ($z = 4.51$).

Simulation algorithm. — The covariance matrix $\hat{\Sigma}$ is estimated once per null model from the same $B = 50,000$ simulated count vectors. The same fixed $\hat{\mu}$ and $\hat{\Sigma}$ are then used to evaluate the Mahalanobis statistic for every null replicate (a fully Monte Carlo reference distribution, not a leave-one-out procedure). The observed count vector is not included in $\hat{\mu}$ or $\hat{\Sigma}$. This procedure is computationally equivalent for all three null models and yields a purely empirical p-value without parametric assumptions.

This approach is inspired by King et al. [9] and related to energy-distance tests [18] and multivariate permutation tests [11], but differs in using Mahalanobis distance from the simulated null mean — requiring no density estimation, pairwise distances, or asymptotic approximations. The p-value is purely empirical. A formal theoretical treatment is planned in future work.

Relationship to Hotelling’s T^2 and max- T . — The test statistic D_M^2 uses the same quadratic form as Hotelling’s T^2 , but the observed value is evaluated against the empirical null distribution of D_M under permutation, not against a parametric reference distribution. Asymptotic arguments

(e.g., $D_M^2 \approx \chi^2(5)$ under Gaussian approximation) serve as heuristic background only; the actual inference is entirely distribution-free. With observed counts as small as 18 (Week-Dilug), the asymptotic approximation is not relied upon; moreover, Hotelling’s T^2 additionally assumes multivariate normality of the test statistic — an assumption violated by small integer counts such as those here — so the parametric form is inappropriate for this application. The max- T procedure, which bases inference on the single most extreme component z-score, is less sensitive than Mahalanobis when multiple patterns are simultaneously displaced from the null; in our data, the two approaches lead to the same substantive conclusion under both null models. The Bonferroni–Holm correction and the joint Mahalanobis test answer different inferential questions: Holm provides family-wise error rate control across five separate component hypotheses; the Mahalanobis statistic tests a single global multivariate null. The joint test does not replace Holm, nor is it uniformly more powerful; it aggregates correlated evidence into one distance measure that is sensitive to departure in any direction from the null centre.

5.4 Non-Significant Patterns

Week, Week-Dilug, and Dilug-in-Dilug were not individually significant. These results do not invalidate these halachic rules, which are established on legal-theological grounds [14]. The non-significance of Dilug-in-Dilug is particularly notable given its large z-score at the sequence level in the joint test: it is the dependence with Haflaga (via the polynomial hierarchy) that elevates its collective contribution to D , even when its individual count is exactly at the null mean. This illustrates why joint and individual tests can differ.

For the Week pattern, it is worth noting that the weekly anchor set $\mathcal{W} = \{22, 29, 36, 43, 50\}$ (in haflaga units) corresponds to cycle lengths $\{21, 28, 35, 42, 49\}$. Since cycles of 35 days or longer are uncommon in the Fehring dataset (9.6% of all cycles; mean $L = 29.3$, $SD = 3.9$), nearly all Week patterns in this sample arise from haflaga values of 22 (cycle length 21) or 29 (cycle length 28). The 28-day cycle is the mode of the dataset and the most biologically common length, so the Week pattern in this dataset effectively captures women with near-28-day cycles who also show the requisite two-consecutive-cycle consistency.

5.5 Clinical and Halachic Implications

The findings carry several implications, all of which should be understood as hypotheses for future study rather than clinical guidance: none of the outcomes discussed below is directly evaluated in the present analysis.

Fertility awareness. — Women with established Haflaga represent the most cycle-regular subgroup in this dataset (within-person SD: 1.54 days vs. 3.17 days for non-vesset women). For such women, backward estimation of the fertile window (approximately 14 days prior to anticipated onset) is most reliable, making Haflaga a candidate marker for calendar-based fertility planning [1]. Whether this marker improves on statistical cycle-length prediction methods will be examined in the companion forecasting study [12].

Halachic family purity compliance. — For observant Jewish women, the vessel system governs the timing of ritual abstinence (*taharat hamishpacha*). The finding that Haflaga is associated with lower observed cycle-length variability motivates evaluation of vessel-derived features as candidate markers of cycle regularity in prospective studies; a direct comparison with established cycle-forecasting methods remains a topic for future work. The non-significance under the within-woman null is consistent with the halachic position that vessel rules codify observed regularity rather than impose temporal structure beyond it. Whether these empirical findings bear on the legal-theological validity of vessel rules is a question outside the scope of this analysis.

Cycle regularity as a health indicator. — Menstrual cycle irregularity is associated with decreased fertility and elevated risk of chronic conditions [1]. The finding that Haflaga marks women with low within-person cycle-length variability ($SD < 1.6$ days) suggests it could function as a simple regularity indicator — a clinical hypothesis to be evaluated in datasets with longitudinal clinical measurements and defined clinical endpoints. Changes in vessel status (loss or gain of Haflaga) may also serve as a signal of disruption to baseline regularity, warranting prospective investigation.

5.6 Scope and Generalizability

Before discussing specific limitations, we emphasise that all conclusions are restricted to this specific dataset of non-Jewish NFP users. The results should not be generalised to the general population or to observant Jewish women without replication. The NFP dataset represents a self-selected group of cycle-aware women, likely more regular than the general population, which may inflate pattern counts. Any inference about the halachic validity of vessel rules in the Jewish context requires datasets collected from the target population. Such validation within an observant Jewish demographic would directly bridge the statistical findings to their halachic application context.

5.7 Limitations

Data quality and coding. — The haflaga convention $H_k = L_k + 1$ is a conventional relabelling; pattern counts are identical whether computed on H_k or L_k *provided the anchor shifts by the same* -1 (verified: Haflaga 26, Week-Dilug 18 under both conventions). Computing on L_k with the unadjusted anchor ($L = 30$ instead of $L = 29$) yields only 12 Week-Dilug events instead of the correct 18. As a sensitivity diagnostic, the raw haflaga distribution was inspected for heaping artefacts at round numbers. Counts decline smoothly from the mode ($H = 29$: 225 cycles) through the Week-Dilug anchor ($H = 30$: 175) to $H = 31$ (141); this pattern is consistent with the known biological mode at 29–30 days rather than digit rounding (Figure 10). This diagnostic is descriptive; a formal statistical test for digit preference is not performed. The NFP dataset records cycle length in whole days but not onset time of day, creating potential misclassification relative to the halachic requirement that all pattern onsets occur in the same onset period (daytime or nighttime). Both upward and downward biases are possible; their relative magnitude cannot be assessed, but the effect is expected to be approximately uniform across women and pattern types. If menstruation onset timing varies systematically with individual physiology, misclassification could differ across women; however, the within-woman null (H_W) conditions on each woman’s observed cycle-length multiset, so any consistent within-woman shift in recorded lengths is preserved identically across observed and permuted sequences, leaving the H_W test unaffected by this form of individual-level misclassification.

Dataset scope. — Calendar dates are absent, precluding analysis of the 10 date-dependent vessel types; two of the 7 cycle-length types (Dilug Chozer Chalila, Haflaga Chozer Chalila) had zero occurrences due to their stringent run-length requirements, leaving the 5 patterns analysed here. The dataset also excludes prodromal physical symptoms (*vessel haguf*): women may have established body-sign vessel patterns entirely invisible to cycle-length analysis.

Population generalizability. — Participants are non-Jewish NFP users [6]: regularly cycling married women aged 18–42. NFP users are a self-selected group likely more cycle-aware and possibly more regular than the general population, introducing a potential upward bias in regularity-dependent patterns such as Haflaga. Several halachic authorities hold that regularities in non-Jewish women may not generalise to Jewish women in practice [16].

Statistical methodology. — The global null models (H_G , H_{iid}) shuffle cycle lengths across women, ignoring within-person autocorrelation; H_W addresses this by conditioning on each woman’s own cycle-length distribution, but tests only the specific exchangeability null that all orderings of a woman’s observed multiset are equally probable — not a general test of serial independence or stationarity. Future work could employ parametric nulls based on fitted state-space models [1, 17]. The pairwise rate-based analysis mitigates the cycle-count exposure confound by normalising event counts by the number of cycles per woman; associations involving rare patterns (Week-Dilug: 18 events; Dilug-in-Dilug: 27 events) should nonetheless be interpreted as exploratory, given the limited number of women establishing those patterns. The Haflaga $\times$ Dilug-in-Dilug association that survives Holm correction is further supported by the polynomial-proximity argument (Section 2.2), providing structural justification beyond the empirical correlation. With a mean of 13.2 cycles per woman, longer patterns such as Dilug-in-Dilug are likely underpowered. As an approximate sensitivity analysis, we computed a minimum detectable count (MDC) for each pattern under H_W at nominal 80% power and $\alpha = .05$ (two-sided), using a normal-theory location-shift approximation:

$$\text{MDC} \approx \mu_W + (z_{0.025} + z_{0.80}) \sigma_W = \mu_W + 2.802 \sigma_W. \quad (2)$$

MDC values are approximate normal-theory benchmarks, not exact power guarantees (Table 5); a simulation-based power analysis is deferred to future work. All five observed counts fall below their MDC, consistent with limited sensitivity at $N = 118$; the joint H_W test is likewise non-significant ($D_M = 2.95$, $p = .121$).

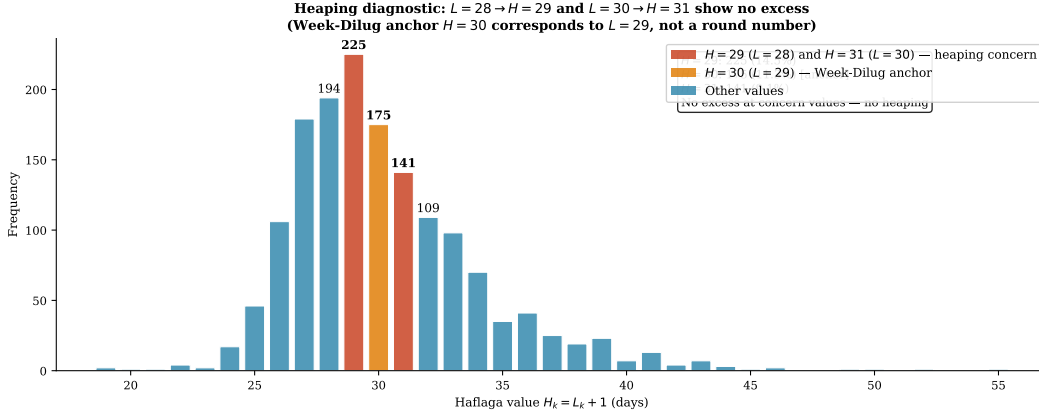


Figure 10: Distribution of haflaga values H_k around $H = 28$ and $H = 30$. The distribution is smooth with no heaping artefact at $H = 30$ ($H = 29$: 225 cycles; $H = 30$: 175; $H = 31$: 141), confirming that the Week-Dilug anchor at $H = 30$ does not reflect a reporting bias.

Table 5: Approximate detectable-count benchmarks under H_W (within-woman permutation null), based on a normal-theory location-shift approximation. μ_W : null mean; σ_W : null SD; 95th pct: empirical 95th percentile of null (rejection threshold at $\alpha = .05$); MDC: minimum detectable count at nominal 80% sensitivity ($\text{MDC} \approx \mu_W + 2.802 \sigma_W$, where $z_{0.025} + z_{0.80} = 1.960 + 0.842 = 2.802$; approximate normal-theory benchmark, not an exact power guarantee); z_{obs} : standardised observed count. All observed counts fall below the MDC, consistent with limited sensitivity at $N = 118$.

Pattern	Obs.	μ_W	σ_W	95th pct	MDC	z_{obs}
Haflaga	26	31.85	4.68	40.0	45	−1.25
Dilug	61	67.07	7.36	79.0	88	−0.83
Week	32	33.81	3.88	40.0	45	−0.47
Week-Dilug	18	20.94	3.22	26.0	30	−0.91
Dilug-in-Dilug	27	39.14	5.86	49.0	56	−2.07

6. Conclusion

This study provides, to our knowledge, the first formal statistical test of whether halachic vessel patterns occur at rates distinguishable from chance in empirical menstrual cycle data (no systematic literature search was performed to confirm priority). The five patterns analysed form a polynomial hierarchy whose structure generates algebraic structural relationships — one a

strict logical containment ($\text{Haflaga-30} \subseteq \text{Week-Dilug}$), one an algebraic approximation ($\text{Haflaga-Dilug-in-Dilug}$ proximity) — explored through exploratory pairwise randomisation-based tests. Three complementary null models were applied: a global permutation null, a multinomial null, and the scientifically sharper within-woman permutation null (H_W).

The primary result is from the within-woman null: under H_W , *no pattern is individually significant* (all Holm-corrected $p \geq .198$; joint $D_M = 2.95$, $p = .121$), and all observed counts fall below their within-woman null means. These findings provide no evidence, under within-woman exchangeability, that temporal ordering within each woman’s cycle sequence creates vessel patterns beyond what her own cycle-length distribution predicts. This is a non-detection of excess under the exchangeability null with acknowledged limited power (all five observed counts fall below the normal-theory MDC; Table 5); it should be interpreted as no detectable evidence of within-woman temporal structure under this null, not as proof of its absence.

Under the global null, Haflaga is significantly overrepresented ($p < .001$ after Holm correction); Dilug is not significant ($p = .087$, Holm adj $p = .346$) and should be treated as hypothesis-generating pending replication. The joint Mahalanobis test confirms significant displacement from the global null ($D_M = 5.10$, $p < .001$; only 2 of 50,000 permutation simulations reached the observed Haflaga count). This global result is best interpreted as a descriptive decomposition: it demonstrates that women who establish Haflaga are drawn from a subpopulation with lower cycle-length variability, a population-level heterogeneity effect that does not, by itself, constitute evidence of within-woman temporal predictability. The primary focused joint test on the three structurally dependent patterns (Haflaga, Week-Dilug, Dilug-in-Dilug) — justified by set-theoretic containment and polynomial-hierarchy proximity prior to data inspection — yields $D_M = 4.52$, $p < .001$. The 5-pattern robustness check ($D_M = 5.10$, $p < .001$) confirms the result is not sensitive to pattern-subset choice. Together, these results suggest that halachic vessel rules function primarily as empirical detectors of individual cycle regularity: the rules identify the subset of women with lower observed cycle-length variability and formalise that regularity as an anticipatory pattern. The joint distance-based permutation test, adapted here for dependent count vectors in sequential menstrual-cycle data (conceptually related to quadratic-form and energy-distance permutation tests; see Section 4.4), provides a

way to aggregate correlated count evidence into a single global multivariate null test. The Holm–Bonferroni correction and the joint Mahalanobis test answer different inferential questions — Holm controls FWER across five separate hypotheses, while the Mahalanobis statistic tests one global null — and the two procedures are complementary rather than competing. This adapted approach may prove useful for composite hypothesis testing in other cyclic biological settings (e.g., multitype rhythm patterns, circadian or hormonal cycle data, episodic clinical event sequences) where structurally related count outcomes are tested jointly.

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References

- [1] Paola Bortot, Guido Masarotto, and Bruno Scarpa. Sequential predictions of menstrual cycle lengths. *Biostatistics*, 11(4):741–755, 2010. doi: 10.1093/biostatistics/kxq020.
- [2] Mitchell D. Creinin, Sharon Keverline, and Leslie A. Meyn. How regular is regular? An analysis of menstrual cycle regularity. *Contraception*, 70(4):289–292, 2004.
- [3] Sonya S. Dasharathy, Sunni L. Mumford, Anna Z. Pollack, Neil J. Perkins, Donald R. Mattison, Jean Wactawski-Wende, and Enrique F. Schisterman. Menstrual bleeding patterns among regularly menstruating women. *American Journal of Epidemiology*, 175(6):536–545, 2012.
- [4] Anthony C. Davison and David V. Hinkley. *Bootstrap Methods and Their Application*. Cambridge University Press, Cambridge, 1997.
- [5] Eugene S. Edgington and Patrick Onghena. *Randomization Tests*. Chapman & Hall/CRC, Boca Raton, FL, 4th edition, 2007.
- [6] Richard J. Fehring, Mary Schneider, and Kathleen Barron. Randomized comparison of two Internet-supported methods of natural family planning. *Contraception*, 88(1):24–30, 2013. doi: 10.1016/j.contraception.2012.10.010.
- [7] Richard J. Fehring, Mary Schneider, and Kathleen Barron. Natural family planning dataset. Marquette University e-Publications, https://epublications.marquette.edu/data_nfp/7/, 2013. Dataset accompanying Fehring et al. (2013).
- [8] Phillip I. Good. *Permutation, Parametric and Bootstrap Tests of Hypotheses*. Springer, New York, 3rd edition, 2005.
- [9] Maxwell L. King, Xibin Zhang, and Muhammad Akram. Hypothesis testing based on a vector of statistics. Working paper, Department of Econometrics and Business Statistics, Monash University. Revised July 1, 2018, 2018.

- [10] Kirsten Münster, Lene Schmidt, and Peter Helm. Length and variation in the menstrual cycle — a cross-sectional study from a Danish county. *British Journal of Obstetrics and Gynaecology*, 99(5):422–429, 1992.
- [11] Fortunato Pesarin and Luigi Salmaso. *Permutation Tests for Complex Data: Theory, Applications and Software*. John Wiley & Sons, Chichester, UK, 2010.
- [12] Dvir Ross. Forecasting efficiency of halachic vessel rules: A comparison with statistical cycle-length prediction methods. Manuscript in preparation.
- [13] Dvir Ross. Reliability of menstrual cycle reporting in surveys with regard to Jewish religious laws. *Functional Differential Equations*, 26(1–2):1–13, 2019. In Hebrew.
- [14] Dvir Ross. Statistical analysis for the religious laws of abstinence before menstruation. *B.D.D*, 2021. Accepted 3 September 2021. In Hebrew.
- [15] Dvir Ross. Fixed-difference menstruation Markov chain. *B.D.D*, 36:77–104, 2022. In Hebrew.
- [16] Dvir Ross. *Probabilistic and Statistical Analysis of the Menstrual Cycle in a Halachic Context*. PhD thesis, Ariel University in Samaria, Ariel, Israel, 2022. In Hebrew. Supervisors: Prof. E. Merzbach, Dr. E. Shmerling, Prof. A. Domoshnitsky. Available at <https://github.com/dvirross/PhD>.
- [17] Laura Symul, Katarzyna Wac, Paula Hillard, and Marcel Salathé. Assessment of menstrual health status and evolution through mobile apps for fertility awareness. *npj Digital Medicine*, 2:64, 2019.
- [18] Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.